

PERTH SWANBOURNE, Australia

WMO: 946140

Lat: 31.9558S

Lon: 115.7619E

Elev: 134

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	43.8	45.7	32.7	27.4	63.4	35.5	30.6	60.6	36.7	61.1	32.2	60.4	8.4	80	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	95.0	67.8	90.9	67.4	87.3	66.9	73.6	83.1	71.9	80.9	70.7	79.6	10.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	115.3	76.9	69.4	108.7	75.2	68.0	103.5	74.7	37.2	83.0	35.7	80.7	34.7	79.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.4	23.5	20.8	DB	39.9	103.5	1.6	2.9	38.8	105.6	37.9	107.3	37.0	109.0	35.8	111.1	(4)
(5)			WB	38.4	75.9	1.7	2.3	37.2	77.5	36.2	78.8	35.3	80.1	34.1	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	74.2	75.0	72.6	67.8	62.8	58.7	57.0	57.5	59.1	62.4	67.5	71.2	(6)	
	DBStd	8.07	5.64	5.86	5.88	4.94	4.15	3.61	3.65	3.53	3.92	4.66	5.39	6.24	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	1121	1	1	5	24	93	189	248	235	181	108	28	8	(9)	
	CDD50	5635	751	700	701	535	396	262	218	231	274	386	524	657	(10)	
	CDD65	1278	287	281	240	109	23	1	0	1	5	29	102	200	(11)	
	CDH74	8978	2186	2274	1709	489	77	1	0	4	25	173	621	1419	(12)	
	CDH80	3285	814	900	615	121	7	0	0	0	5	46	215	562	(13)	
(14) Wind	WSAvg	11.1	12.0	11.5	11.1	9.9	10.1	10.5	11.0	10.7	11.2	11.0	11.8	12.0	(14)	
(15) Precipitation	PrecAvg	28.8	0.4	0.5	0.9	1.5	3.7	5.6	5.7	4.9	2.9	1.4	0.9	0.4	(15)	
	PrecMax	36.5	4.2	4.5	2.9	4.3	6.6	10.1	9.1	7.6	6.1	3.9	2.5	1.8	(16)	
	PrecMin	20.9	0.0	0.0	0.0	0.3	1.7	1.9	2.5	1.2	0.9	0.1	0.0	0.0	(17)	
	PrecStd	4.2	0.9	0.9	0.8	1.0	1.3	2.2	1.7	1.3	1.2	0.9	0.7	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.2	98.8	97.9	89.0	80.4	72.9	70.6	72.6	76.8	85.8	93.0	97.9	(19)	
		MCWB	68.7	69.8	69.0	64.9	60.7	57.6	57.1	58.4	59.4	62.8	64.5	67.5	(20)	
	2%	DB	92.9	93.7	90.8	83.3	75.9	69.8	67.3	68.8	71.6	78.8	86.1	91.5	(21)	
		MCWB	68.2	68.8	67.3	63.8	61.4	58.3	57.7	58.6	59.5	61.0	63.9	66.6	(22)	
	5%	DB	88.1	89.6	86.5	79.2	72.9	67.6	65.5	66.4	68.6	73.8	80.8	86.3	(23)	
		MCWB	67.9	68.8	66.2	64.1	60.8	57.8	57.3	58.0	59.5	61.3	62.8	65.9	(24)	
	10%	DB	83.9	85.5	82.4	75.9	70.4	65.7	63.7	64.5	66.2	70.6	76.4	81.5	(25)	
		MCWB	67.6	68.2	65.9	63.8	60.7	57.8	56.8	57.5	58.6	60.9	63.1	65.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	76.7	73.8	71.1	68.3	64.5	63.4	64.3	65.5	67.3	69.8	72.7	(27)	
		MCDB	84.7	87.7	80.7	75.8	71.9	67.0	65.5	67.2	69.0	74.7	77.1	84.4	(28)	
	2%	WB	73.1	74.2	72.0	69.3	66.1	62.7	61.2	62.0	63.0	65.3	68.0	70.8	(29)	
		MCDB	83.1	83.5	79.4	74.8	70.1	65.8	63.9	65.1	67.2	72.1	75.7	80.8	(30)	
	5%	WB	71.4	72.2	70.6	67.9	64.4	61.1	59.7	60.3	61.4	63.8	66.7	69.4	(31)	
		MCDB	80.5	81.9	78.6	73.8	69.0	64.6	62.8	63.9	66.0	70.0	74.6	79.1	(32)	
	10%	WB	70.1	70.7	69.3	66.3	62.7	59.6	58.3	58.7	59.8	62.3	65.2	68.0	(33)	
		MCDB	79.3	80.4	78.0	72.7	68.1	63.5	61.9	62.7	64.8	68.7	73.4	77.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.1	17.6	16.0	15.3	13.9	13.4	14.0	14.2	15.5	16.1	16.7	(35)	
		MCDBR	24.7	24.6	23.8	20.6	18.5	16.2	15.2	16.4	18.2	21.5	24.3	24.6	(36)	
	5% WB	MCWBR	10.9	10.6	10.1	10.3	10.1	9.5	9.4	10.0	10.6	11.3	11.4	11.1	(37)	
		MCWBR	18.2	18.1	16.6	14.5	13.6	12.0	11.3	12.4	14.5	16.4	17.2	18.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.339	0.339	0.324	0.317	0.307	0.302	0.300	0.308	0.326	0.326	0.327	0.328	(40)	
		taud	2.592	2.606	2.643	2.646	2.646	2.640	2.641	2.614	2.541	2.558	2.579	2.598	(41)	
	Ebn at Noon		317	311	305	291	279	272	278	290	299	312	319	322	(42)	
		Edh at Noon	33	32	29	26	23	22	23	26	31	33	33	33	(43)	
(44) All-Sky Solar Radiation	RadAvg		2599	2343	1934	1411	1051	856	907	1183	1583	2063	2459	2663	(44)	
	RadStd		117	79	94	82	68	59	57	59	92	85	101	105	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.23	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page